

5.5

(a) independent variable $X_t = t$

$$\begin{aligned}\sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{\left(\sum_{t=1}^T t\right)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T} \\&= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} = \frac{2T(T+1)(2T+1)}{12} - \frac{3T(T+1)^2}{12} \\&= \frac{T(T+1)[2(2T+1) - 3(T+1)]}{12} = \frac{T(T+1)(4T+2-3T-3)}{12} \\&= \frac{T(T+1)(T-1)}{12}\end{aligned}$$

$$\Rightarrow \text{var}(b_2|X) = \frac{\sigma^2}{\frac{T(T+1)(T-1)}{12}} = \frac{12\sigma^2}{T(T+1)(T-1)}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{var}(b_2|X) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T+1)(T-1)} = 0$$

$\therefore$ the least squares estimator b_2 is consistent.

(b) independent variable $X_t = 0.5^t$

$$\sum_{t=1}^T 0.5^t = \frac{0.5(1-0.5^T)}{1-0.5} = 1-0.5^T$$

$$\sum_{t=1}^T (0.5^t)^2 = \sum_{t=1}^T 0.25^t = \frac{0.25(1-0.25^T)}{1-0.25} = \frac{1}{3}(1-0.25^T)$$

$$\Rightarrow \sum_{t=1}^T (X_t - \bar{X})^2 = \frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}$$

$$\Rightarrow \text{var}(b_2|X) = \frac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{var}(b_2|X) = \frac{\sigma^2}{\frac{1}{3} \times 1 - 0} = 3\sigma^2 \neq 0$$

$\therefore$ the least squares estimator b_2 is not consistent.

(c)

In part (a), increasing T provides more and more information about β_2 and the relationship between y and x .

In part (b), $x_t = 0.5^t$ becomes zero after the initial few periods, and increasing T does not provide any more information about β_2 .

5.6

(a)

$$H_0: \beta_2 = 0$$

$$\alpha = 0.05$$

$$H_1: \beta_2 \neq 0$$

$$N = 63$$

$$\text{test statistic } t = \frac{b_2 - 0}{\text{se}(b_2)} \sim t_{(N-3)}$$

$$\Rightarrow t^* = \frac{3 - 0}{\sqrt{4}} = 1.5$$

$$t_c = t_{0.025}(60) = 2.000$$

$$RR = \{t: |t| > 2.000\}$$

$$\because |t^*| = 1.5 < 2.000$$

$\therefore$ we do not reject H_0 .

(b)

$$H_0: \beta_1 + 2\beta_2 = 5$$

$$H_1: \beta_1 + 2\beta_2 \neq 5$$

$$\text{test statistic } t = \frac{(b_1 + 2b_2) - 5}{\text{se}(b_1 + 2b_2)} \sim t_{(N-3)}$$

$$\Rightarrow t^* = \frac{2 + 6 - 5}{\sqrt{3 + 4 \times 4 + 4 \times (-2)}} = \frac{3}{\sqrt{11}} = 0.9045$$

$$t_c = t_{0.025}(60) = 2.000$$

$$RR: \{t: |t| > 2.000\}$$

$$\because |t^*| = 0.9045 < 2.000$$

$\therefore$ we do not reject H_0 .

(c)

$$H_0: \beta_1 - \beta_2 + \beta_3 = 4$$

$$H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$$

$$\text{test statistic } t = \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} \sim t_{(N-3)}$$

$$\Rightarrow t^* = \frac{2 - 3 + (-1) - 4}{\sqrt{3 + 4 + 3 - 2 \times (-2) + 2 \times 1 - 2 \times 0}} = \frac{-6}{\sqrt{16}} = -1.5$$

$$t_c = t_{0.025}(60) = 2.000$$

$$RR: \{t: |t| > 2.000\}$$

$$\because |t^*| = 1.5 < 2.000$$

$\therefore$ we do not reject H_0 .

5.20

(a)

$$E(\hat{\beta}_{2, \text{mean}} | X) = E\left[\frac{(\bar{y}_2 - \bar{y}_1)}{(\bar{x}_2 - \bar{x}_1)} \middle| X\right] = \frac{1}{(\bar{x}_2 - \bar{x}_1)} E[(\bar{y}_2 - \bar{y}_1) | X]$$

$$\begin{aligned} E[(\bar{y}_2 - \bar{y}_1) | X] &= E(\bar{y}_2 | X) - E(\bar{y}_1 | X) \\ &= (\beta_1 + \beta_2 \bar{x}_2) - (\beta_1 + \beta_2 \bar{x}_1) \\ &= \beta_2 (\bar{x}_2 - \bar{x}_1) \end{aligned}$$

$$\Rightarrow E(\hat{\beta}_{2, \text{mean}} | X) = \frac{1}{(\bar{x}_2 - \bar{x}_1)} \times \beta_2 (\bar{x}_2 - \bar{x}_1) = \beta_2$$

$$\begin{aligned} \Rightarrow E(\hat{\beta}_{2, \text{mean}}) &= E_X [E(\hat{\beta}_{2, \text{mean}} | X)] \\ &= E_X (\beta_2) = \beta_2 \end{aligned}$$

$\therefore \hat{\beta}_{2, \text{mean}}$ is an unbiased estimator.

(b)

$$\begin{aligned}\text{var}(\hat{\beta}_{2,\text{mean}}|X) &= \left[\frac{1}{(\bar{x}_2 - \bar{x}_1)} \right]^2 \cdot \text{var}[(\bar{y}_2 - \bar{y}_1)|X] \\ &= \left[\frac{1}{(\bar{x}_2 - \bar{x}_1)} \right]^2 \cdot [\text{var}(\bar{y}_2|X) + \text{var}(\bar{y}_1|X)]\end{aligned}$$

$$\begin{aligned}\Rightarrow \text{var}(\bar{y}_1|X) &= \text{var}\left(\frac{1}{\frac{N}{2}} \sum_{i=1}^{\frac{N}{2}} y_i \mid X\right) \\ &= \left(\frac{1}{\frac{N}{2}}\right)^2 \cdot \left[\sum_{i=1}^{\frac{N}{2}} \text{var}(y_i|X)\right] \\ &= \left(\frac{4}{N^2}\right) \cdot \left(\frac{N}{2} \sigma^2\right) \\ &= \frac{2\sigma^2}{N} = \text{var}(\bar{y}_2|X)\end{aligned}$$

$$\begin{aligned}\therefore \text{var}(\hat{\beta}_{2,\text{mean}}|X) &= \left[\frac{1}{(\bar{x}_2 - \bar{x}_1)} \right]^2 \cdot \left(\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N} \right) \\ &= \left[\frac{1}{(\bar{x}_2 - \bar{x}_1)} \right]^2 \cdot \left(\frac{4\sigma^2}{N} \right) \\ &= \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\end{aligned}$$

(c)

$$\text{var}(\hat{\beta}_2, \text{mean}) = E[\text{var}(\hat{\beta}_2, \text{mean} | X)] + \text{var}[E(\hat{\beta}_2, \text{mean} | X)]$$

$\because E(\hat{\beta}_2, \text{mean} | X) = \beta_2$ is a constant.

$$\therefore \text{var}[E(\hat{\beta}_2, \text{mean} | X)] = 0$$

$$\Rightarrow \text{var}(\hat{\beta}_2, \text{mean}) = E[\text{var}(\hat{\beta}_2, \text{mean} | X)] = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = 4E\left[\frac{\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right]$$

(d)
Condition: $\lim_{N \rightarrow \infty} \text{var}(\hat{\beta}_2, \text{mean}) = \lim_{N \rightarrow \infty} 4E\left[\frac{\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = 0$

$$\Rightarrow (\bar{x}_2 - \bar{x}_1)^2 \neq 0$$

$$\Rightarrow \bar{x}_2 \neq \bar{x}_1$$

$\therefore \hat{\beta}_2, \text{mean}$ will be a consistent estimator for β_2 if $\bar{x}_2 \neq \bar{x}_1$.

(e)

(i)

	N=100	N=500	N=1000	N=5000
$\text{var}(b_2 X)$	1.016706	0.237710	0.120282	0.024196
$\text{var}(\hat{\beta}_2, \text{mean} X)$	1.320880	0.323183	0.161417	0.032206

(ii)

	N=100	N=500	N=1000	N=5000
$E[(s_x^2)^{-1}]$	0.101671	0.118855	0.120282	0.120982
$E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right]$	0.132088	0.161592	0.161417	0.161031

(f)

In part (i), holding sample size constant, $\text{var}(b_2|X)$ is always smaller than $\text{var}(\hat{\beta}_{2, \text{mean}}|X)$. As sample size increases, $\text{var}(b_2|X)$ and $\text{var}(\hat{\beta}_{2, \text{mean}}|X)$ both become smaller.

In part (ii), holding sample size constant, $E[(S_x^2)^{-1}]$ is always smaller than $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}]$. As sample size increases, $E[(S_x^2)^{-1}]$ and $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}]$ are higher but not significant.

Because $\text{var}(\hat{\beta}_{2, \text{mean}}|X)$ approaches 0 as $N \rightarrow \infty$ and $\hat{\beta}_{2, \text{mean}}$ has been shown to be unbiased, $\hat{\beta}_{2, \text{mean}}$ is a consistent estimator.

(g)

Show that $E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$:

$$\text{var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12}$$

$$\text{As } N \rightarrow \infty, S_x^2 \xrightarrow{P} \text{var}(X) = \frac{100}{12}$$

$$\Rightarrow \text{By Continuous Mapping Theorem: } \text{plim}(S_x^2)^{-1} = \frac{1}{\frac{100}{12}} = \frac{12}{100} = 0.12$$

Show that $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}] \xrightarrow{P} 0.16$:

the first half sample is equivalent to drawing from $U(0, 5)$

the second half sample is equivalent to drawing from $U(5, 10)$

$$\Rightarrow \text{plim}(\bar{x}_1) = E[X | X \in U(0, 5)] = \frac{0+5}{2} = 2.5$$

$$\text{plim}(\bar{x}_2) = E[X | X \in U(5, 10)] = \frac{5+10}{2} = 7.5$$

$$\text{plim}(\bar{x}_2 - \bar{x}_1) = 7.5 - 2.5 = 5$$

$$\text{By Continuous Mapping Theorem: } \text{plim}\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4}{(5)^2} = \frac{4}{25} = 0.16$$

(h)

These two halves of samples can be consider as two independent random samples drawn from $U(0,10)$.

$$\Rightarrow \text{plim}(\bar{X}_1) = E(X) = \frac{0+10}{2} = 5$$

$$\text{plim}(\bar{X}_2) = E(X) = \frac{0+10}{2} = 5$$

$$\text{plim}(\bar{X}_2 - \bar{X}_1) = 5 - 5 = 0$$

By Continuous Mapping Theorem :

$$\text{plim} \hat{\beta}_{2, \text{mean}} = \text{plim} \frac{\bar{Y}_2 - \bar{Y}_1}{\bar{X}_2 - \bar{X}_1} = \text{plim} \frac{\bar{Y}_2 - \bar{Y}_1}{0} \neq \beta_2$$

$\therefore \hat{\beta}_{2, \text{mean}}$ is not a consistent estimator.